Run ID: u574_1791578260_5c7282

Adaptive Probe-Guided Genetic Algorithm (APGGS) — Experiment Report

Generated: 2026-10-10 02:24:46
Engine Version: v4.2

1. Executive Summary & Dataset Metadata

Dataset Name	U574	**Cities / Nodes**	574
Edge Weight Type	EUC_2D	**Known Best Optimum**	36,905
Final Global Best Cost	36,905.00	**Optimality Gap**	+0.0000%
Best Found Generation	15	**Completed Generations**	15
Stagnation Streak	0 / 30	**Stopping Condition**	30 gens without improvement
Experiment Status	COMPLETED	**Total Execution Time**	1054.29 s
Stop Reason	0.0% gap achieved (Known optimum found)	**Random Seed**	1

2. Initial High-Quality Route Construction

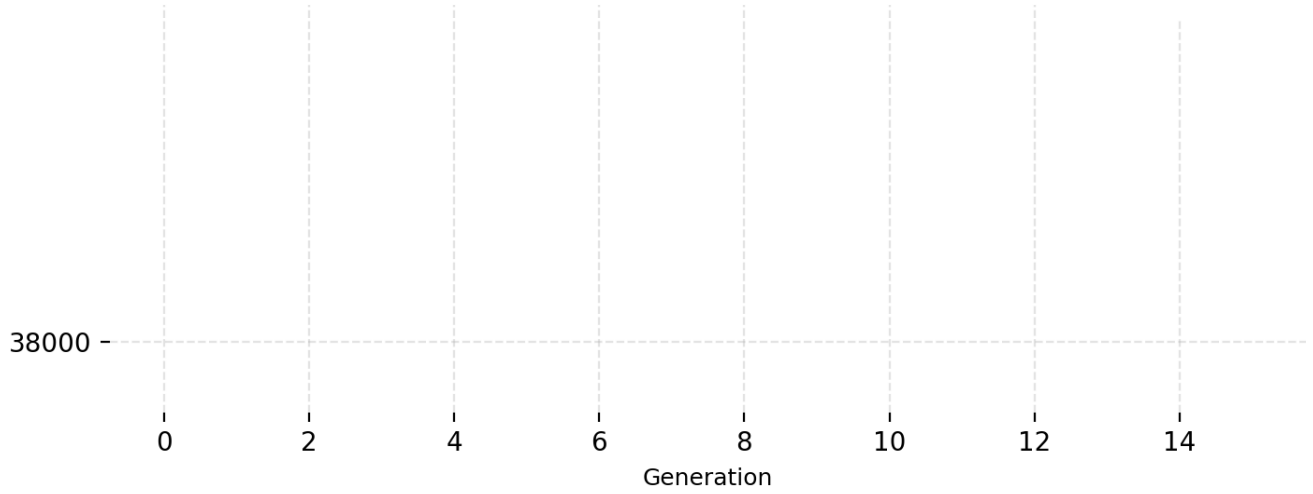
Construction Stage	Resulting Tour Cost	Improvement vs Prev	Duration (s)
Stage 1: Greedy Nearest-Neighbor Tour	45,043.00	Baseline	0.4154
Stage 2: 2-Opt Local Search Improvement	44,803.00	-240.00	0.0004
Stage 3: 2-Opt Deep Refinement Seed	44,685.00	-118.00	0.0010
Stage 4: Initial 70-Individual Population (NN+Greedy+Random)	44,685.00	Population Seeded	28.5700

3. Genetic Algorithm & Adaptive Pipeline Configuration

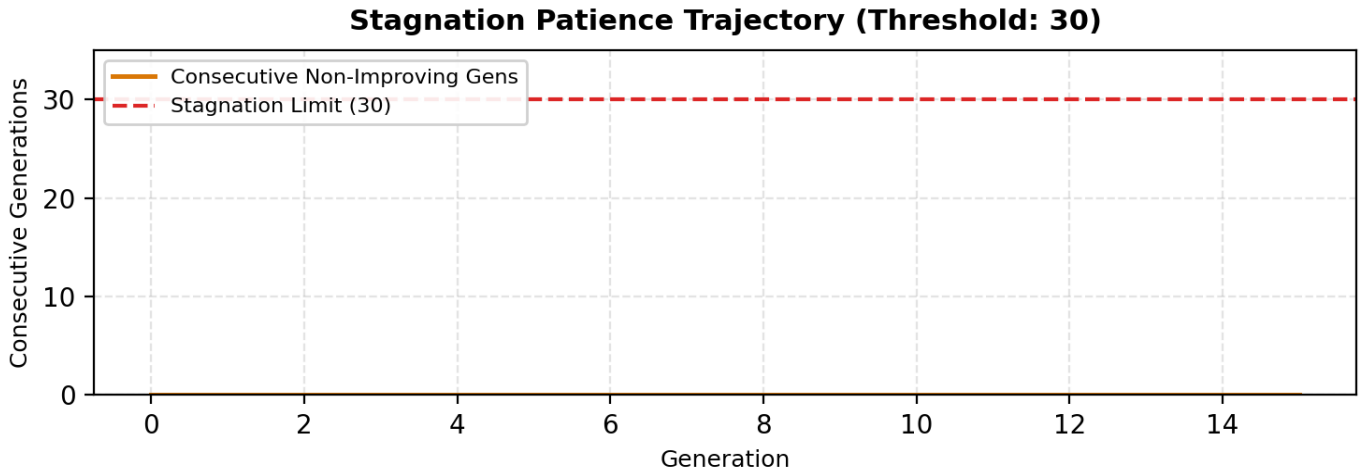
Parameter	Configured Value	Parameter	Configured Value
Population Size	70	Mutation Rate	0.20
Nearest-Neighbor Seeds	17	Swap Fraction	0.02
Greedy Edge Seeds	11	Segment Crossover Prob	0.40
Random Tour Seeds	42	Offspring 2-Opt Prob	0.35
K-Nearest Neighbors	15	Adaptive Elites Count	14
Tournament Size	4	PMG Memory Decay	0.92
Selective LKH Cadence	Every 15 gens	Stagnation Window	15
Selective LKH Individuals	3	Stagnation Patience Limit	30 consecutive gens

4. Research Visualizations & Trajectory Curves

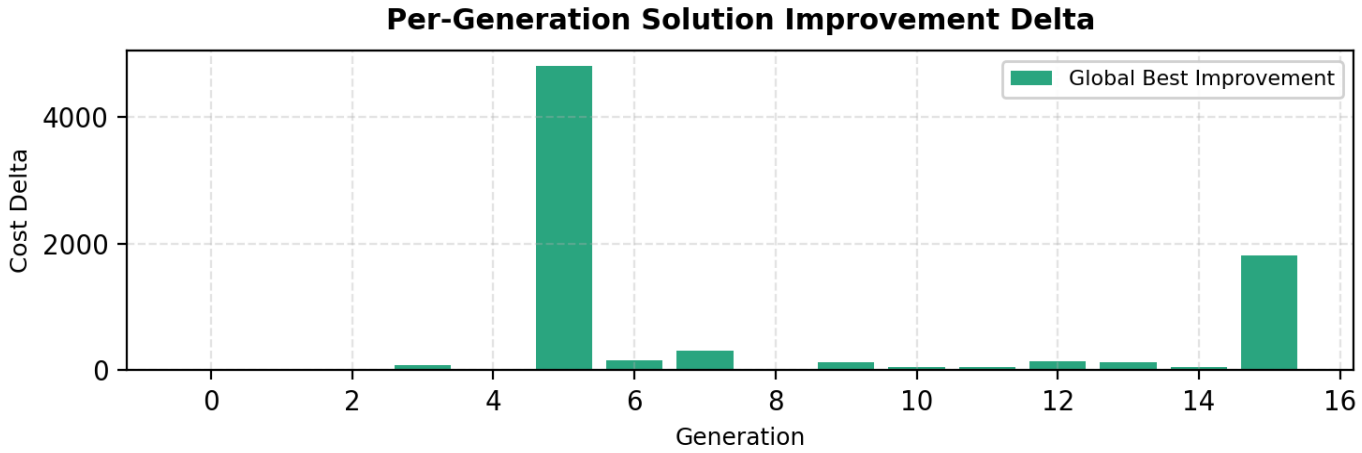
Convergence Trajectory: Global Best Cost vs Generation

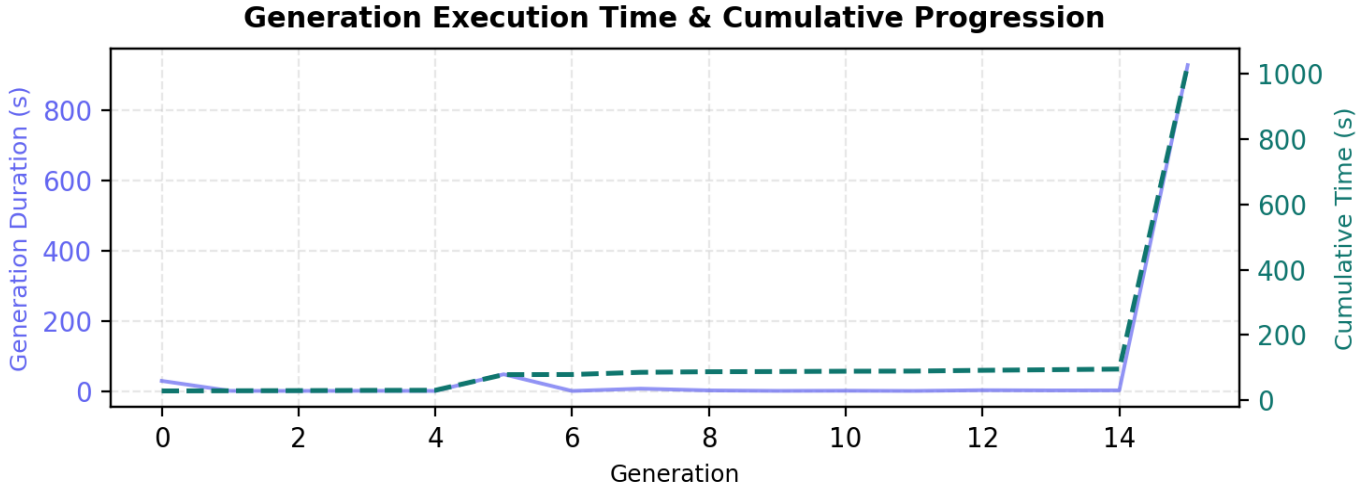


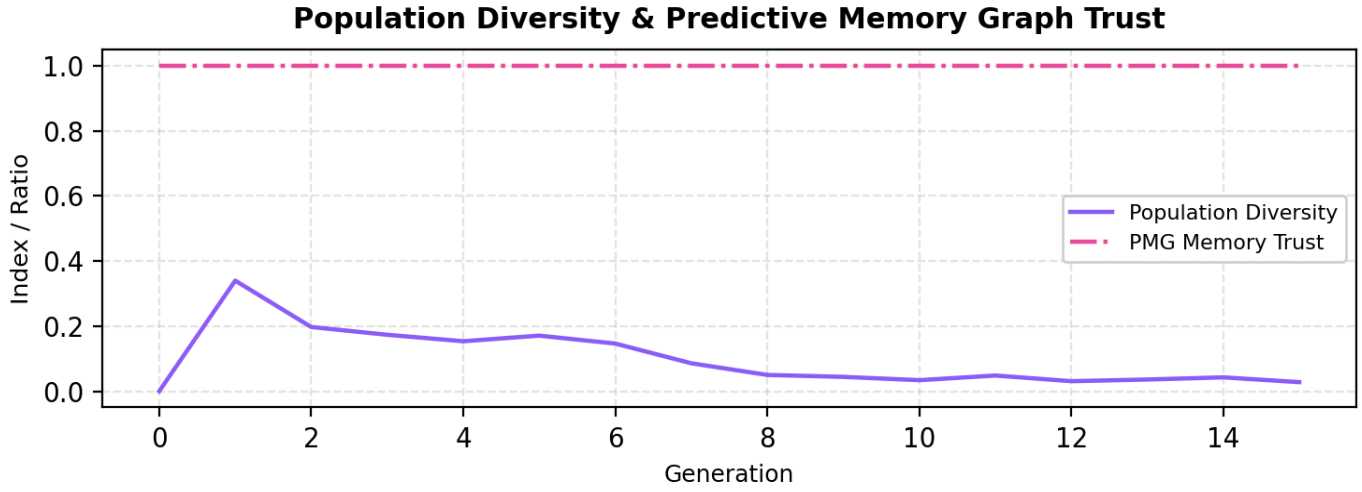
Stagnation Counter Progression (Target: 30 Consecutive Non-Improving Gens)



Per-Generation Cost Improvement & Runtime Progression







5. Evolution Highlights (Global-Best Discoveries)

Gen	Global Best Cost	Cost Reduction	Gap (%)	Operator / Method	PMG Trust	Non-Imp Streak	Duration (s)
0	44,685.00	0.00	+21.0812%	Greedy + 2Opt + 2Opt + Selective LKH Seed	1.00	0	28.9867
1	44,669.00	16.00	+21.0378%	Elite + weak_edge_surgery	1.00	0	0.5613
2	44,655.00	14.00	+20.9999%	Elite + two_opt	1.00	0	0.4160
3	44,573.00	82.00	+20.7777%	OX Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	0.7607
4	44,558.00	15.00	+20.7370%	Elite + two_opt	1.00	0	0.3723
5	39,758.00	4,800.00	+7.7307%	OX Crossover + predictive_or_opt	1.00	0	47.8326
6	39,595.00	163.00	+7.2890%	OX Crossover + Fast 2-Opt + two_opt	1.00	0	0.2780
7	39,289.00	306.00	+6.4598%	Elite + predictive_or_opt	1.00	0	6.7484
8	39,280.00	9.00	+6.4354%	Segment Crossover + Fast 2-Opt	1.00	0	1.7831
9	39,147.00	133.00	+6.0751%	Segment Crossover + Fast 2-Opt + two_opt	1.00	0	0.5634
10	39,092.00	55.00	+5.9260%	Elite + weak_edge_surgery	1.00	0	0.9692
11	39,044.00	48.00	+5.7960%	OX Crossover + weak_edge_surgery	1.00	0	0.4103
12	38,899.00	145.00	+5.4031%	Segment Crossover + Fast 2-Opt + predictive_or_opt	1.00	0	2.3868
13	38,773.00	126.00	+5.0616%	OX Crossover + Fast 2-Opt + weak_edge_surgery	1.00	0	1.8199
14	38,722.00	51.00	+4.9235%	OX Crossover + Fast 2-Opt	1.00	0	1.9308
15	36,905.00	1,817.00	+0.0000%	OX Crossover + Fast 2-Opt + predictive_or_opt + Selective LKH	1.00	0	929.4837

6. Final Discovered Tour & Solution Vector

Optimized Tour Length: 36,905.00 | **Optimality Gap:** +0.0000% | **Discovered at Generation:** 15

Permutation Sequence Preview:

```
0 → 567 → 568 → 569 → 566 → 565 → 564 → 563 → 1 → 2 → 3 → 4 → 5 → 562 → 561 → 560 → 559 → 558 → 554 → 555
→ 556 → 557 → 6 → 7 → 8 → 9 → 10 → 11 → 553 → 552 → 551 → 550 → 549 → 548 → 547 → 546 → 545 → 12 → 13 →
544 → ... [534 additional cities] → 0
```

7. Execution Event Log & Diagnostics

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• Initialized u574 (574 cities). Initial best cost: 44,685.00
• Gen 1: Discovered new global-best 44,669.00 (-16.00)
• Gen 2: Discovered new global-best 44,655.00 (-14.00)
• Gen 3: Discovered new global-best 44,573.00 (-82.00)
• Gen 4: Discovered new global-best 44,558.00 (-15.00)
• Gen 5: Discovered new global-best 39,758.00 (-4,800.00)
• Gen 6: Discovered new global-best 39,595.00 (-163.00)
• Gen 7: Discovered new global-best 39,289.00 (-306.00)
• Gen 8: Discovered new global-best 39,280.00 (-9.00)
• Gen 9: Discovered new global-best 39,147.00 (-133.00)
• Gen 10: Discovered new global-best 39,092.00 (-55.00)
• Gen 11: Discovered new global-best 39,044.00 (-48.00)
• Gen 12: Discovered new global-best 38,899.00 (-145.00)
• Gen 13: Discovered new global-best 38,773.00 (-126.00)
• Gen 14: Discovered new global-best 38,722.00 (-51.00)
• Gen 15: Discovered new global-best 36,905.00 (-1,817.00)
• Gen 15: 0.0% optimality gap achieved. Experiment completed.
```

8. Artifact Manifest & Research Reproducibility

All raw experimental records have been saved to runs/u574_1791578260_5c7282/ including summary.json, generation_history.jsonl, best_route.csv, and report.pdf.